

1. Declare variables for a Flipkart product: `productName` (as a string), `price` (float), and `rating` (double). Assign sample values and print each variable with its data type.

```
2.    #include <stdio.h>
3.
4.    int main()
5.    {
6.        // Variable declarations
7.        char productName[] = "Samsung Galaxy S24";
8.        float price = 74999.99f;
9.        double rating = 4.8;
10.
11.        // Printing values with their data types
12.        printf("Product Name (String): %s\n", productName);
13.        printf("Price (Float): %.2f\n", price);
14.        printf("Rating (Double): %.11f\n", rating);
15.
16.        return 0;
17.    }
```

2. Create a constant variable to store the GST rate (for example, 18%) and use it to calculate the final price of a Zomato order with a given base price.  
**Constraint:** The GST rate must not be changeable after its initial assignment.

```
    #include <stdio.h>

int main()
{
    // Constant variable for GST rate (18%)
    const float GST_RATE = 18.0;

    // Base price of the Zomato order
    float basePrice = 500.0;

    // Calculate GST amount and final price
    float gstAmount = (basePrice * GST_RATE) / 100;
    float finalPrice = basePrice + gstAmount;

    // Display the result
    printf("Base Price: Rs. %.2f\n", basePrice);
    printf("GST Rate: %.2f%\n", GST_RATE);
    printf("GST Amount: Rs. %.2f\n", gstAmount);
    printf("Final Price: Rs. %.2f\n", finalPrice);

    return 0;
}
```

3. Write a program that stores your favorite Spotify playlist's name (string), total number of songs (int), and average song duration in minutes (float). Print all values in a single formatted sentence.

```
#include <stdio.h>

int main()
{
    // Variable declarations
    char playlistName[] = "Chill Vibes";
    int totalSongs = 35;
    float averageDuration = 3.8;

    // Print all values in one formatted sentence
    printf("My favorite Spotify playlist is \"%s\", which contains %d songs\n",
           playlistName, totalSongs, averageDuration);

    return 0;
}
```

4. Identify and correct the invalid variable names from this list: 1stPlayer, player\_age, \$score, total-marks, userName.<br><br><em><strong>Hint:</strong> Apply standard variable naming rules for your programming language.</em>

| Original Variable Name | Valid/Invalid                                          | Correct Variable Name     |
|------------------------|--------------------------------------------------------|---------------------------|
| 1stPlayer              | <input type="checkbox"/> Invalid (starts with a digit) | firstPlayer OR player1    |
| player_age             | <input type="checkbox"/> Valid                         | player_age                |
| \$score                | <input type="checkbox"/> Invalid (contains \$)         | score                     |
| total-marks            | <input type="checkbox"/> Invalid (contains -)          | total_marks OR totalMarks |
| userName               | <input type="checkbox"/> Valid                         | userName                  |